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CS427 HW6

1) A one round Feistel cipher cannot be a secure PRP because of only half of the of the plain text is changed each time. IE there is no diffusion which is a property that makes ciphertext appear random.
    a. A distinguisher would be to divide a plan text into parts A and B, and compute C = (L,K) If C = L then F is not a PRP because the ciphertext is only a function of half of the plain text and it not random

       If C != L then F may not be a PRP. Either way this distinguisher only focuses on the structure of a Feistel transformation and it not secure.

2) A distinguisher for a 2 round keyed Feistal Cipher would something like as follows:
    a. Pick two plain texts A and B where B is fixed as the all zero string. Generate the cipertexts f(a) and f(b) then computer the XOR of f(a) and f(b) IE C = f(a) XOR (fb)

       Then one will see that C is the XOR outputs of the round functions after the first round IE C = f1(a) XOR f1(b)
       Because of the fact that Fiestel transformation only operates on half of the input and that the outputs are XORed together in the next round this means the difference between A and B in this case are fixed. IE the XOR outputs of the two rounds is a fixed output so this makes it distinguishable from a random permutation and this violating the definition of a secure PRP.
3) A PRF is defined as secure when the probability that the attackers result is zero minus the probably that the attackers result is one is negligible.

   F is a secure PRF with λ-bit outputs, meaning that for any adversary A, the probability of guessing the output of F for a random key k and input r is at most 2^-λ.

   Since G is a PRG with stretch `, it is infeasible for any adversary to determine the output of G for a given input.

   Therefore, it is infeasible for any adversary to determine the output of F' for a random key k and input r, which satisfies the unpredictability requirement.

   As a result, we can conclude that F' is a secure PRF with outputs of length λ +`.